

习题课

空间解析几何

一、内容小结

二、实例分析



一、内容小结

1. 空间直线与平面的方程

空间平面

一般式 $Ax + By + Cz + D = 0 \quad (A^2 + B^2 + C^2 \neq 0)$

点法式 $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

截距式 $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

点: (x_0, y_0, z_0)
法向量: $\vec{n} = (A, B, C)$

三点式
$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$



空间直线

一般式
$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0 \\ A_2x + B_2y + C_2z + D_2 = 0 \end{cases}$$

对称式
$$\frac{x - x_0}{m} = \frac{y - y_0}{n} = \frac{z - z_0}{p}$$

参数式
$$\begin{cases} x = x_0 + mt \\ y = y_0 + nt \\ z = z_0 + pt \end{cases}$$

(x_0, y_0, z_0) 为直线上一点;

$\vec{s} = (m, n, p)$ 为直线的方向向量.



2. 线面之间的相互关系

面与面的关系

平面 $\Pi_1 : A_1x + B_1y + C_1z + D_1 = 0$, $\vec{n}_1 = (A_1, B_1, C_1)$

平面 $\Pi_2 : A_2x + B_2y + C_2z + D_2 = 0$, $\vec{n}_2 = (A_2, B_2, C_2)$

垂直: $\vec{n}_1 \cdot \vec{n}_2 = 0 \iff A_1A_2 + B_1B_2 + C_1C_2 = 0$

平行: $\vec{n}_1 \times \vec{n}_2 = \vec{0} \iff \frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$

夹角公式: $\cos \varphi = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$



线与线的关系

$$\text{直线 } L_1: \frac{x-x_1}{m_1} = \frac{y-y_1}{n_1} = \frac{z-z_1}{p_1}, \quad \vec{s}_1 = (m_1, n_1, p_1)$$

$$\text{直线 } L_2: \frac{x-x_2}{m_2} = \frac{y-y_2}{n_2} = \frac{z-z_2}{p_2}, \quad \vec{s}_2 = (m_2, n_2, p_2)$$

$$\text{垂直: } \vec{s}_1 \cdot \vec{s}_2 = 0 \iff m_1 m_2 + n_1 n_2 + p_1 p_2 = 0$$

$$\text{平行: } \vec{s}_1 \times \vec{s}_2 = \vec{0} \iff \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{p_1}{p_2}$$

$$\text{夹角公式: } \cos \varphi = \frac{|\vec{s}_1 \cdot \vec{s}_2|}{|\vec{s}_1| |\vec{s}_2|}$$



面与线间的关系

平面: $Ax + By + Cz + D = 0$, $\vec{n} = (A, B, C)$

直线: $\frac{x - x_0}{m} = \frac{y - y_0}{n} = \frac{z - z_0}{p}$, $\vec{s} = (m, n, p)$

垂直: $\vec{s} \times \vec{n} = \vec{0} \iff \frac{m}{A} = \frac{n}{B} = \frac{p}{C}$

平行: $\vec{s} \cdot \vec{n} = 0 \iff mA + nB + pC = 0$

夹角公式: $\sin \varphi = \frac{|\vec{s} \cdot \vec{n}|}{|\vec{s}| |\vec{n}|}$



3. 相关的几个问题

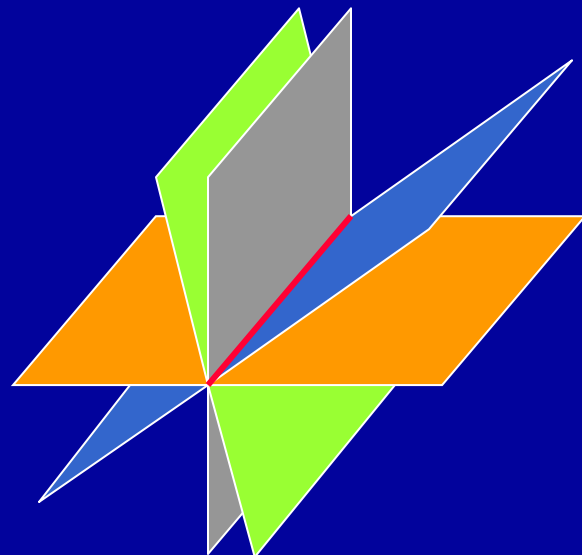
(1) 过直线

$$L: \begin{cases} A_1x + B_1y + C_1z + D_1 = 0 \\ A_2x + B_2y + C_2z + D_2 = 0 \end{cases}$$

的平面束 方程

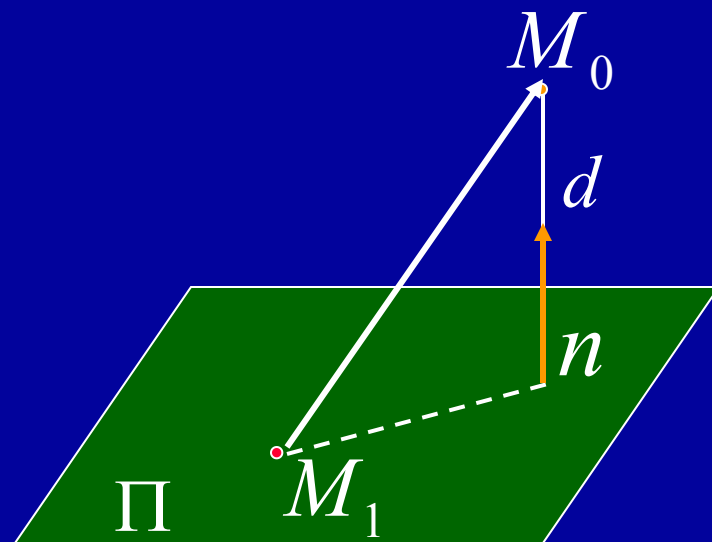
$$\lambda_1(A_1x + B_1y + C_1z + D_1) + \lambda_2(A_2x + B_2y + C_2z + D_2) = 0$$

(λ_1, λ_2 不全为0)



(2) 点 $M_0(x_0, y_0, z_0)$ 到平面 $\Pi: Ax + By + Cz + D = 0$ 的距离为

$$\begin{aligned} d &= \frac{|\overrightarrow{M_1 M_0} \cdot \vec{n}|}{|\vec{n}|} \\ &= \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}} \end{aligned}$$



$$M_1(x_1, y_1, z_1)$$

$$\begin{aligned} \overrightarrow{M_1 M_0} \cdot \vec{n} &= A(x_0 - x_1) + B(y_0 - y_1) + C(z_0 - z_1) \\ &= Ax_0 + By_0 + Cz_0 + D \end{aligned}$$

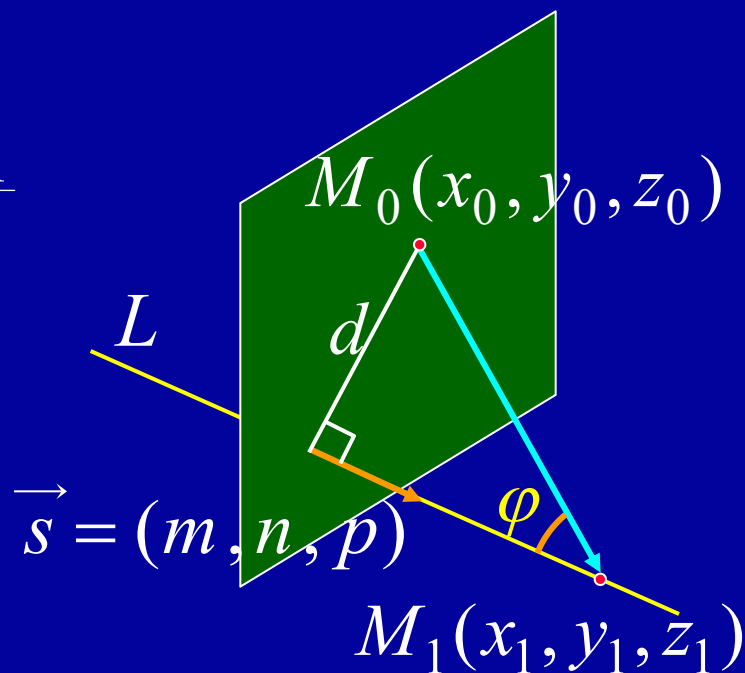


(3) 点 $M_0(x_0, y_0, z_0)$ 到直线

$$L: \frac{x-x_1}{m} = \frac{y-y_1}{n} = \frac{z-z_1}{p}$$

的距离为

$$d = \frac{|\overrightarrow{M_0M_1} \times \vec{s}|}{|\vec{s}|}$$



$$= \frac{1}{\sqrt{m^2 + n^2 + p^2}} \left| \begin{array}{ccc} \vec{i} & \vec{j} & \vec{k} \\ x_1 - x_0 & y_1 - y_0 & z_1 - z_0 \\ m & n & p \end{array} \right|$$



二、实例分析

例1. 求与两平面 $x - 4z = 3$ 和 $2x - y - 5z = 1$ 的交线平行, 且过点 $(-3, 2, 5)$ 的直线方程.

提示: 所求直线的方向向量可取为

$$\vec{s} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -4 \\ 2 & -1 & -5 \end{vmatrix} = (-4, -3, -1)$$

利用点向式可得方程

$$\frac{x+3}{4} = \frac{y-2}{3} = \frac{z-5}{1}$$



例2. 求直线 $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{2}$ 与平面

$$2x + y + z - 6 = 0$$

$= t$

的交点 .

提示: 化直线方程为参数方程

$$\begin{cases} x = 2 + t \\ y = 3 + t \\ z = 4 + 2t \end{cases}$$

代入平面方程得 $t = -1$

从而确定交点为 $(1, 2, 2)$.



例3. 求过点 $(2, 1, 3)$ 且与直线 $\frac{x+1}{3} = \frac{y-1}{2} = \frac{z}{-1}$ 垂直相交的直线方程.

提示: 先求二直线交点 P . 过已知点且垂直于已知直线的平面的法向量为 $(3, 2, -1)$, 故其方程为

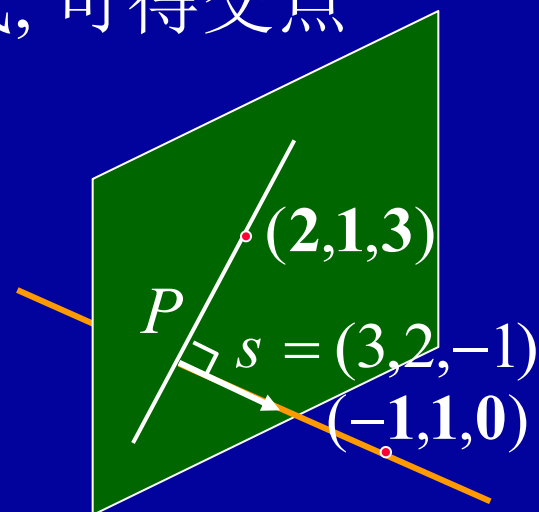
$$3(x-2) + 2(y-1) - (z-3) = 0 \quad ①$$

化已知直线方程为参数方程, 代入 ①式, 可得交点

$$P\left(\frac{2}{7}, \frac{13}{7}, \frac{-3}{7}\right)$$

最后利用两点式得所求直线方程

$$\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-3}{4}$$



例4. 求直线 $\begin{cases} x + y - z - 1 = 0 \\ x - y + z + 1 = 0 \end{cases}$ 在平面 $x + y + z = 0$

上的投影直线方程.

提示: 过已知直线的平面束方程

$$x + y - z - 1 + \lambda(x - y + z + 1) = 0$$

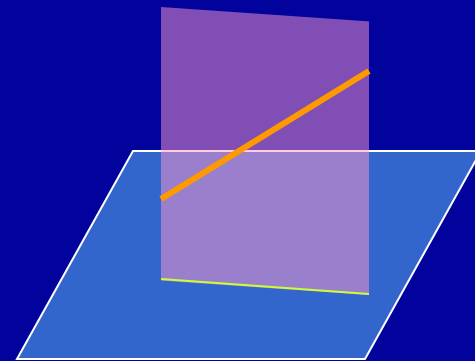
即 $(1 + \lambda)x + (1 - \lambda)y + (-1 + \lambda)z + (-1 + \lambda) = 0$

从中**选择** λ 使其与已知平面垂直, 故应有:

$$(1 + \lambda) \cdot 1 + (1 - \lambda) \cdot 1 + (-1 + \lambda) \cdot 1 = 0$$

得 $\lambda = -1$, 从而得投影直线方程

$$\begin{cases} y - z - 1 = 0 & \longleftarrow \text{这是投影平面} \\ x + y + z = 0 & \longleftarrow \text{这是给定的平面} \end{cases}$$



例5. 设一平面平行于已知直线 $\begin{cases} 2x - z = 0 \\ x + y - z + 5 = 0 \end{cases}$
且垂直于已知平面 $7x - y + 4z - 3 = 0$, 求该平面法线的
的方向余弦.

提示: 已知平面的法向量 $\vec{n}_1 = (7, -1, 4)$
求出已知直线的方向向量 $\vec{s} = (1, 1, 2)$

取所求平面的法向量

$$\vec{n} = \vec{s} \times \vec{n}_1 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 2 \\ 7 & -1 & 4 \end{vmatrix} = 2(3, 5, -4)$$

所求为 $\cos \alpha = \frac{3}{\sqrt{50}}, \cos \beta = \frac{5}{\sqrt{50}}, \cos \gamma = \frac{-4}{\sqrt{50}}$



例6. 求过直线 $L: \begin{cases} x+5y+z=0 \\ x-z+4=0 \end{cases}$ 且与平面 $x-4y-8z+12=0$ 夹成 $\frac{\pi}{4}$ 角的平面方程.

提示: 过直线 L 的平面束方程

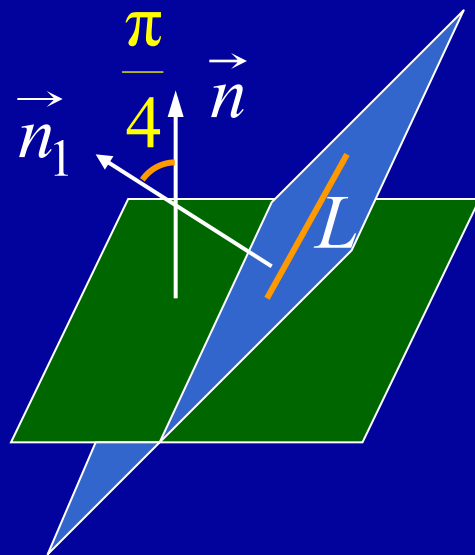
$$(1+\lambda)x + 5y + (1-\lambda)z + 4\lambda = 0$$

其法向量为 $\vec{n}_1 = (1+\lambda, 5, 1-\lambda)$.

已知平面的法向量为 $\vec{n} = (1, -4, -8)$

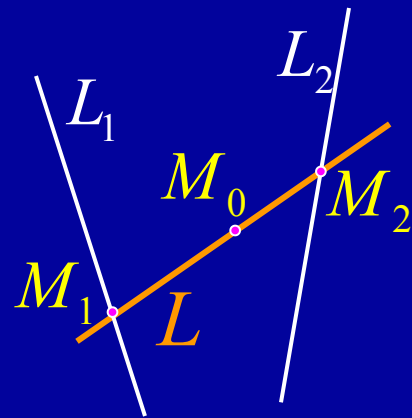
$$\text{选择 } \lambda \text{ 使 } \cos \frac{\pi}{4} = \frac{|\vec{n} \cdot \vec{n}_1|}{\|\vec{n}\| \|\vec{n}_1\|} \implies \lambda = -\frac{3}{4}$$

从而得所求平面方程 $x + 20y + 7z - 12 = 0$.



例7. 求过点 $M_0(1,1,1)$ 且与两直线 $L_1: \begin{cases} y = 2x \\ z = x - 1 \end{cases}$,
 $L_2: \begin{cases} y = 3x - 4 \\ z = 2x - 1 \end{cases}$ 都相交的直线 L .

提示: 思路: 先求交点 M_1, M_2 ;
再写直线方程.



将 L_1, L_2 的方程化为参数方程

$$L_1: \begin{cases} x = t \\ y = 2t \\ z = t - 1 \end{cases}, \quad L_2: \begin{cases} x = t \\ y = 3t - 4 \\ z = 2t - 1 \end{cases}$$

设 L 与它们的交点分别为

$$M_1(t_1, 2t_1, t_1 - 1), \quad M_2(t_2, 3t_2 - 4, 2t_2 - 1).$$



M_0, M_1, M_2 三点共线

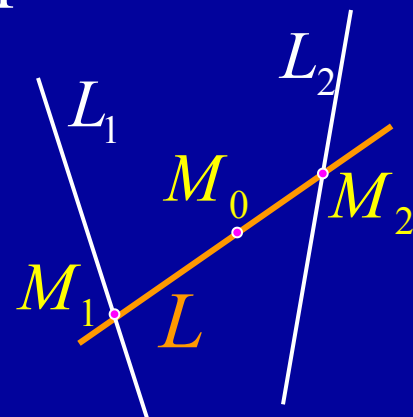
$$\Rightarrow \overrightarrow{M_0M_1} \parallel \overrightarrow{M_0M_2}$$

$$\Rightarrow \frac{t_1 - 1}{t_2 - 1} = \frac{2t_1 - 1}{(3t_2 - 4) - 1} = \frac{(t_1 - 1) - 1}{(2t_2 - 1) - 1}$$

$$\Rightarrow t_1 = 0, t_2 = 2$$

$$\Rightarrow M_1 = (0, 0, -1), M_2 = (2, 2, 3)$$

$$\Rightarrow L: \frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{2}$$



$$M_0(1, 1, 1), \quad M_1(t_1, 2t_1, t_1 - 1), \quad M_2(t_2, 3t_2 - 4, 2t_2 - 1)$$



例8. 直线 $L: \frac{x-1}{0} = \frac{y}{1} = \frac{z}{1}$ 绕 z 轴旋转一周, 求此旋转曲面的方程.

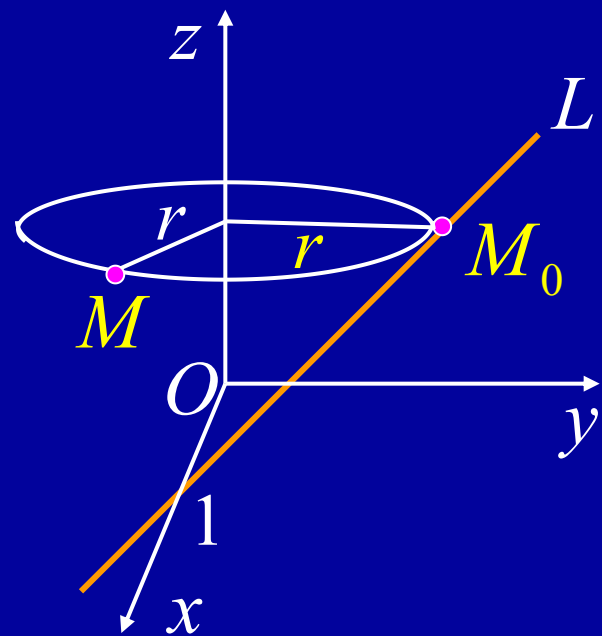
提示: 在 L 上任取一点 $M_0(1, y_0, z_0)$

设 $M(x, y, z)$ 为 M_0 绕 z 轴旋转轨迹上任一点, 则有

$$\begin{cases} z = z_0 = y_0 \\ x^2 + y^2 = 1 + y_0^2 \end{cases}$$

将 $y_0 = z$ 代入第二方程,
得旋转曲面方程

$$x^2 + y^2 - z^2 = 1$$



思考与练习

P50 题21 画出下列各曲面所围图形:

(1) 抛物柱面 $2y^2 = x$, 平面 $z = 0$ 及 $\frac{x}{4} + \frac{y}{2} + \frac{z}{2} = 1$;

(2) 抛物柱面 $x^2 = 1 - z$, 平面 $y = 0$, $z = 0$ 及 $x + y = 1$;

(4) 旋转抛物面 $x^2 + y^2 = z$, 柱面 $y^2 = x$, 平面 $z = 0$
及 $x = 1$.



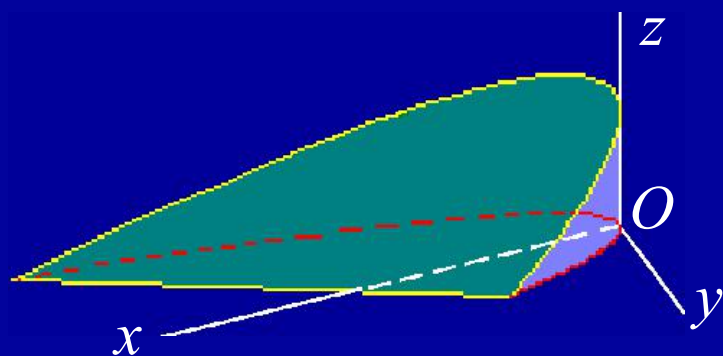
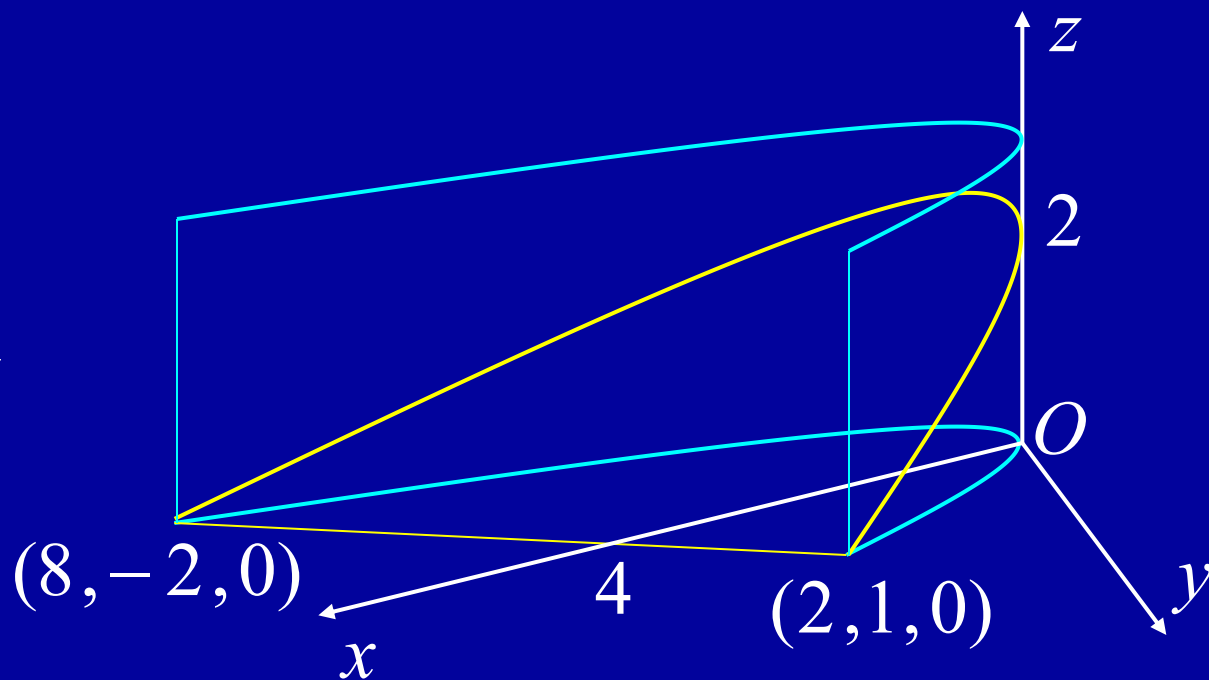
解答:

P50 题21(1)

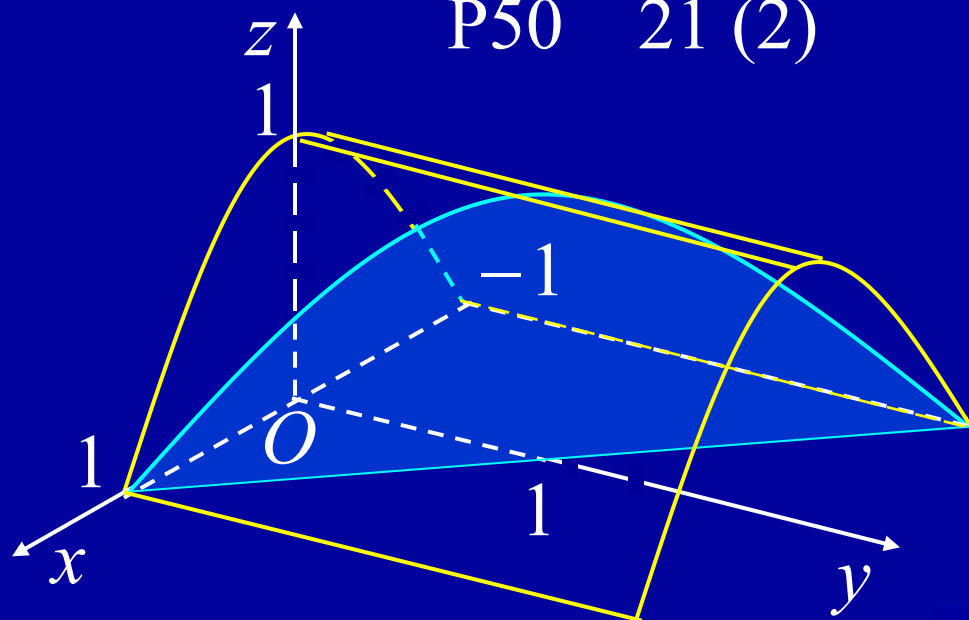
$$2y^2 = x$$

$$\frac{x}{4} + \frac{y}{2} + \frac{z}{2} = 1$$

$$z = 0$$



P50 21 (2)

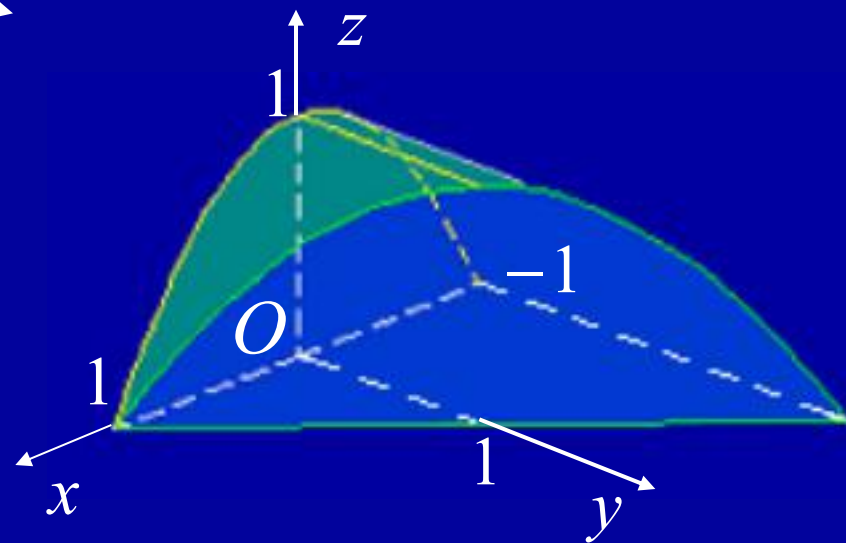


$$x^2 = 1 - z$$

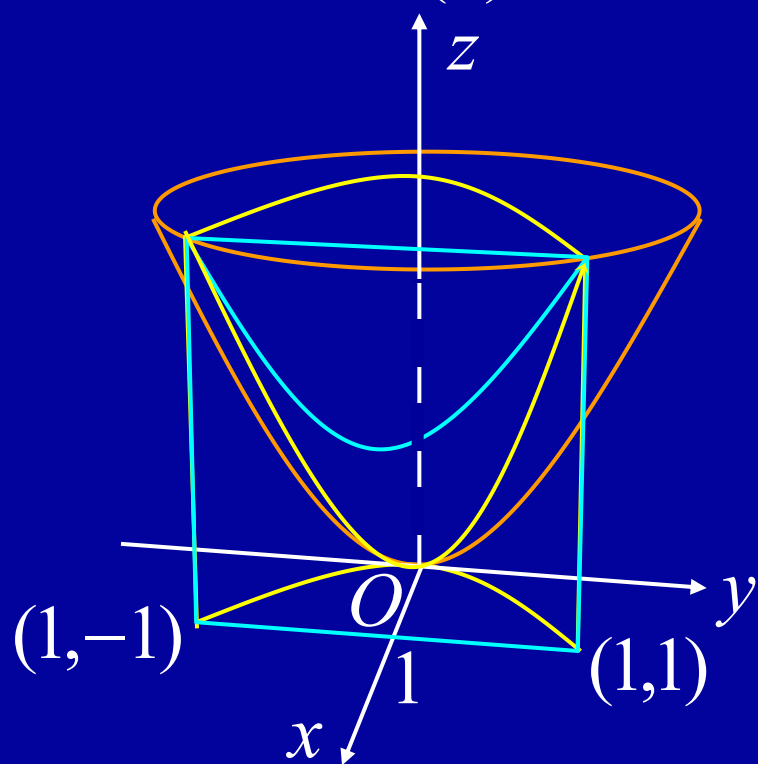
$$y = 0 \quad xOz \text{面}$$

$$z = 0 \quad xOy \text{面}$$

$$x + y = 1$$



P50 21(4)

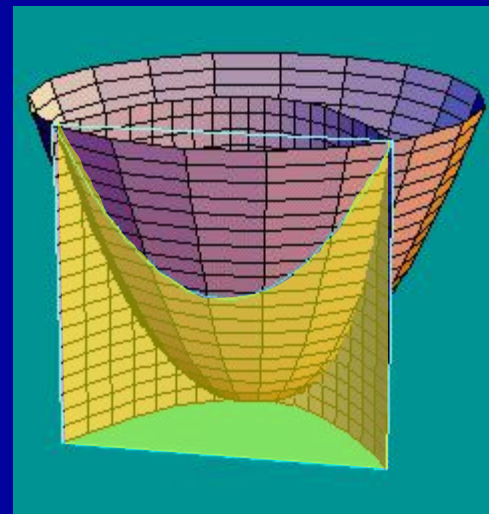


$$x^2 + y^2 = z$$

$$x = 1$$

$$y^2 = x$$

$$z = 0$$



作业

P48 11; 13; 15

P50 15; 16; 17; 18



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